situations, the neighboring spins' Hamiltonian is proportional to $\vec{\sigma} \cdot \vec{\tau}$. If that happens to be the case, then measuring $\vec{\sigma} \cdot \vec{\tau}$ is equivalent to measuring the energy of the atomic pair. Measuring this energy is a single measurement of the composite operator and does not entail measuring the individual components of either spin.

Exercise 6.9: Prove that the four vectors $|sing\rangle$, $|T_1\rangle$, $|T\rangle_2$, and $|T_3\rangle$ are eigenvectors of $\vec{\sigma} \cdot \vec{\tau}$. What are their eigenvalues?

Take a look at your results from this last exercise. Do you see why one of these state-vectors is called the singlet, while the other three are called triplets? The reason is that if you look at their relation to the operator $\vec{\sigma} \cdot \vec{\tau}$, the singlet is an eigenvector with one eigenvalue, and the triplets are all eigenvectors with a different *degenerate* eigenvalue.

Here is a good exercise that combines the concept of entanglement with the concepts of time and change from Lecture 4. Use it to review the ideas of unitary time evolution and the meaning of the Hamiltonian.

Exercise 6.10: A system of two spins has the Hamiltonian

$$\mathbf{H} = \frac{\omega}{2} \vec{\sigma} \cdot \vec{\tau}.$$

What are the possible energies of the system, and what are the eigenvectors of the Hamiltonian?

Suppose the system starts in the state $|uu\rangle$. What is the state at any later time? Answer the same question for initial states of $|ud\rangle$, $|du\rangle$, and $|dd\rangle$.